BETOBULLET

42 HP · EURORACK FORMAT
FOR VCV RACK 2



HARMONION

HARMONIC CV PROCESSOR

USER MANUAL · v2.0.0 · October 2026

CONTENTS

1. Introduction
2. Quickstart
3. Panel overview
4. General features
5. Control voltages and gates
6. Octave switches and sections
7. Chords and arpeggio
8. Scales and modes
9. Chord progression sequencer
10. Specifications

Harmonion puts one global harmony (root, degree and mode) at the centre of your system: three quantizers, a four-voice chord generator, an arpeggiator and a chord progression sequencer all follow it.

1. INTRODUCTION

Harmonion is designed for musicians who are not content with a fixed scale. Its sections share one harmonic context: at any time all three quantizer channels, the chords and the arpeggiator play in the same key and chord, yet each section chooses which notes of that harmony it may use. Change the degree and every voice in your rack moves to the new chord.

Harmonion reproduces the behaviour of the ACL Sinfonion (firmware 1.0), reimplemented from the public user manual. The chapters below follow the structure of that manual; its detailed descriptions apply to Harmonion.

THE SECTIONS

- **Channel 1, 2 and 3**: three independent quantizers.
- **Chords**: four pitch outputs for voiced chords.
- **Arpeggio**: an arpeggiator over the selected chord notes.
- **Chord progression sequencer**: 16 songs × 4 parts × 32 steps of harmony.

INSTALLATION

Copy `Harmonion-2.x-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

2. QUICKSTART

SET ALL TO START

1. Turn ROOT to C, DEGREE to 1, MODE to Chords, TRANS to 0.
2. Select ROOT, 3RD, 5TH and 7TH on Channel 2 and on the Chords section.

USING THE QUANTIZER

Patch a slow LFO into Channel 2 IN and Channel 2 OUT to a VCO. The VCO now plays only the notes whose buttons are lit. Turn DEGREE: the same buttons now mean the notes of another chord.

PLAYING CHORDS

Patch Chords OUT 1–4 into four VCOs (or one polyphonic VCO). Choose up to four notes with the chord buttons; INV sets the inversion, SPREAD opens the voicing.

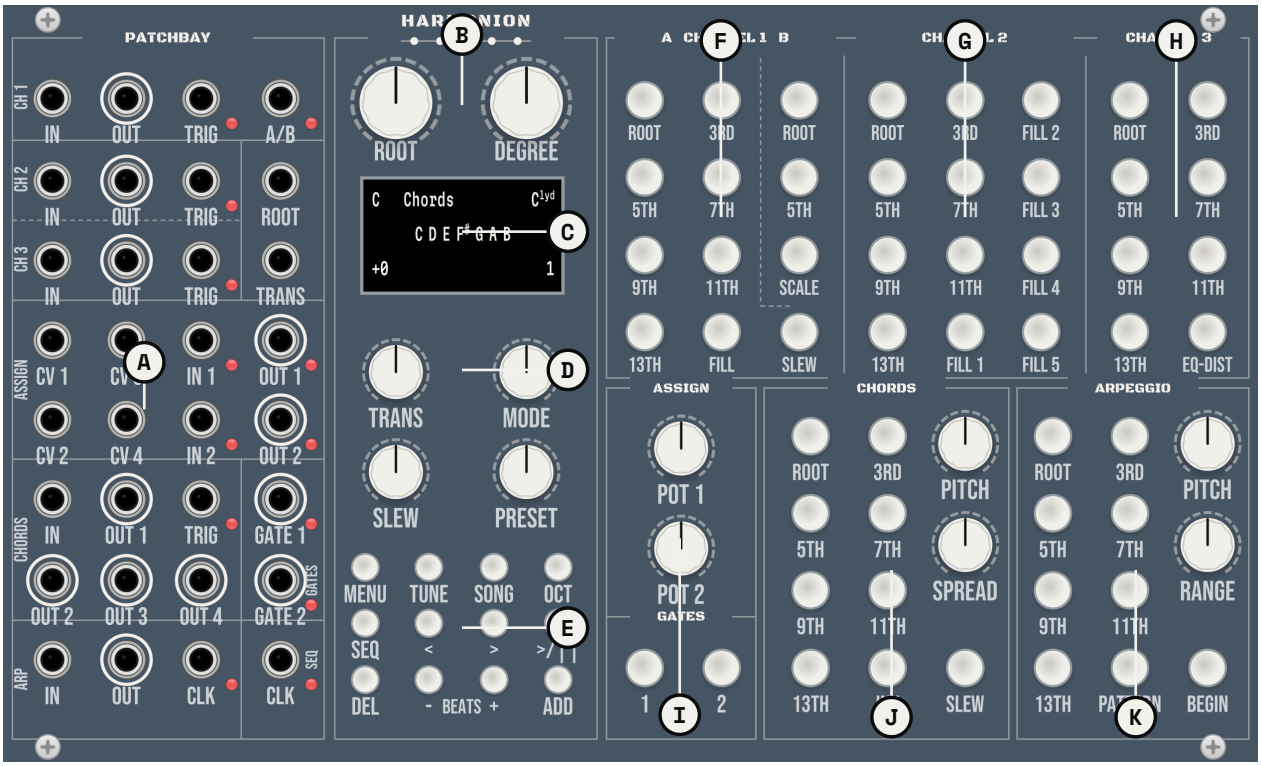
THE ARPEGGIATOR

Patch a clock into Arp CLK and Arp OUT into a VCO. Choose notes and a PATTERN; PITCH sets the lowest note and RANGE the span.

CHORD PROGRESSION SEQUENCER

Press SEQ, patch a clock into the sequencer CLK and press ▶/II. Each step stores ROOT, DEGREE, MODE, the buttons and the octave switches; edit with ← → ADD DEL − +.

3. PANEL OVERVIEW



A **Patchbay**
IN, OUT and TRIG of channels 1–3, A/B, ROOT, TRANS, CV 1–4, IN 1/2, OUT 1/2, chords, gates, arpeggio and sequencer clocks.

B **ROOT and DEGREE**
The key and the chord within it.

C **Display**
Harmony, chord name, transposition and preset or song; menus and pages.

D **TRANS, MODE, SLEW, PRESET, POT 1 / 2**
Global transposition, mode, slew time, button presets and two assignable knobs.

E **Small buttons**
MENU, TUNE, SONG, OCT, SEQ, ←, →, ▶/II, DEL, −, +, ADD.

F **Channel 1**
Note buttons A and B, FILL, A/B and SLEW.

G **Channel 2**
Note buttons and FILL 1–5 (single note mode).

H **Channel 3**
Note buttons and EQ-DIST.

I **Assign / Gates**
POT 1 and POT 2 area and the GATE 1 / 2 buttons.

J **Chords**
Note buttons, INV, SLEW, PITCH and SPREAD.

K **Arpeggio**
Note buttons, PATTERN, BEGIN, PITCH and RANGE.

4. GENERAL FEATURES

THE MENU

MENU opens the configuration; ← → (or − +) select an item and the value; MENU leaves. Items, in order:

ITEM	VALUES
CV 1–4	28 functions (see Control voltages)
POT 1 / POT 2	the same 28 functions
IN 1 / IN 2	20 gate functions
OUT 1 / OUT 2	14 gate functions
Clock Divider	1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 16, 24, 32, 48, 64
Arpeggio Clock	16 ratios from 1/4 to 4/1
Intonation	Equal, Just, Quarter-comma Meantone
Slew Curve	linear, exponential
Trigger Time Window	window in which a trigger samples the input
Arp Trigger Delay	delay of the arpeggio trigger
Pitch Bend Delay	None, Short, Medium, Long
Ch 1 A/B	Off, via Jack, B on Beat 1, Each Odd Beat, Each Even Beat, B on 1st Beat of Step, B on Clock 1, Each Odd Clock, Each Even Clock
Ch 2 Buttons	scale buttons or single note
Display	Scale Tones, Scale Structure, Chord Structure, Octave Switches
Reset To Defaults	resets menu and matrices; keeps songs and presets

THE DISPLAY

- **Top left**: root note and mode. **Top right**: chord or scale name.
- **Bottom left**: global transposition (−12…+12). **Bottom right**: button preset 1–16, or song and part (e.g. 4A) in sequenced mode, with beats (circles) and steps (boxes) in the centre.
- **Centre**: scale tones, scale structure, chord structure or octave switches, as set by the Display menu item.

TUNING MODE, PRESETS AND SLEW

TUNE makes all pitch outputs play the same reference note for tuning VCOs; ← → step through octaves. **PRESET** recalls one of 16 button presets (52 buttons each). **SLEW** sets the glide time; SLEW buttons per section enable it; Slew Curve chooses linear or exponential glide.

5. CONTROL VOLTAGES AND GATES

CV 1–4 AND POT 1/2 FUNCTIONS

Transpose, Chaotic Note Shuffle, Chaotic Detune, Harmonic Shift, Button Preset, Slew Time, Voltmeter, Root, Degree, Mode, Chan 1/2/3 Pre Trans, Arpeggio Pre Trans, Chan 1/2/3 Pitch Bend, Chords Pitch Bend, Chords Spread, Chords Voicing, Arpeggio Clock, Arpeggio Randomize, Arpeggio Range, Song, Song Part 1V/Part, Song + Part, Sequence Step. A pot set to the same function as a CV attenuates that CV, as on the hardware.

IN 1 / IN 2 FUNCTIONS

Sync Slave, Chan 1/2/3 Slew, Chords Slew, Quantize Ch 1/2/3, Quantize Chords, Chords Shuffle, Arpeggio Reset, Seq. Song Reset, Seq. Part Reset, Step Forward/Backward, Part Forward/Backward, Tuning Mode, Seq. Play/Pause.

OUT 1 / OUT 2 FUNCTIONS

Sync Master, Chan 1/2/3 Autogate, Ch 2 Manual Gate, Arpeggio Beat, Sequencer Clock, Sequencer Beat, Song Step, Song Reset, Tuning Mode Active, Ch 1 A/B State, Chord Change.

6. OCTAVE SWITCHES AND SECTIONS

OCT opens a second page for the 52 buttons:

- **Octave switches**: −3…+3 octaves for channels 1–3, chords and arpeggio.
- **Input follows root**: the input is shifted with the global root.
- **Bypass**: three states, from normal quantizing to a soft bypass and a true bypass.
- **Input scaling**: 1, 2, 3 or 5 V per octave at the input.
- **Slew detection**: detects glides at the input and reproduces them at the output.
- **Input / output matrices**: a 4×4 input mixing matrix and a 4×4 output switching matrix for channels 1–3 and the arpeggio.

THE QUANTIZERS IN DETAIL

- **Input trigger**: with TRIG patched, a channel samples and holds; otherwise it quantizes continuously.
- **Autogates**: a gate on every note change (OUT 1/2 functions).
- **FILL buttons**: the notes outside the current scale.
- **Channel 1 A/B**: two button sets switched by jack or rhythm.
- **Channel 2 single note**: every button plays its note directly, with a manual gate.
- **Channel 3 EQ-DIST**: equidistant quantizing over the lit notes only.

7. CHORDS AND ARPEGGIO

CHORD GENERATOR

Up to four selected notes go to OUT 1–4 (lowest to highest). INV chooses Free, Root position, 1st, 2nd or 3rd inversion; SPREAD opens the voicing; PITCH transposes; Chords Voicing by CV selects one of 98 voicings; slash chords keep a fixed bass.

ARPEGGIATOR

PATTERN: Up, Down, Up/Down, Down/Up, Octaves, Baroque, Random, Brownian. BEGIN: Next Note, Root, 3rd, 5th, 7th, 9th, 11th, 13th (where the arpeggio restarts on a chord change). PITCH sets the lowest note, RANGE the span; Arpeggio Clock divides or multiplies the clock and Arp Trigger Delay lets the harmony settle before the note.

8. SCALES AND MODES

ROOT, DEGREE and MODE define the current scale of 6 to 8 notes, mapped to the buttons ROOT, 3RD, 5TH, 7TH, 9TH, 11TH, 13TH; the remaining semitones live on FILL 1–5.

MODE	CHARACTER
Chords	the most common chords of popular music and jazz
Chords2	a second set of chords and colours
Slash chords	chords over a fixed bass
Major, Minor	diatonic church modes on each degree
Mel.Minor, Har.Minor, Har.Major, Db.H.Maj	the modes of melodic minor, harmonic minor, harmonic major and double harmonic major

9. CHORD PROGRESSION SEQUENCER

16 songs, each with parts A–D of up to 32 steps; every step lasts 1–128 beats and stores ROOT, DEGREE, MODE, the buttons and the octave switches.

- **SEQ** switches sequenced mode on; **▶/II** play/pause, double press for **freeze**.
- **← →** select the step, **ADD / DEL** insert or remove steps, **− +** set the beats.
- **SONG** opens the song page: jump to steps, choose parts, live mode.
- Each part has its own Clock Divider.

SYNCING TWO OR MORE HARMONIONS

Set OUT 1 of one Harmonion to **Sync Master** and IN 1 of the others to **Sync Slave**, then patch them: a 16-channel polyphonic cable carries the harmony from master to slaves.

VCV SPECIFICS

Two-button combinations of the hardware also exist in the context menu (pattern, begin, inversion, factory reset, calibration). Gates are 10 V by default (5 V available). Everything is saved in the patch.

10. SPECIFICATIONS

ITEM	VALUE
Width	42 HP
Buttons / knobs	52 large, 12 small / 12
Pitch outputs	8 (Ch 1–3, Arp, Chords 1–4)
Gate outputs	4 (OUT 1, OUT 2, GATE 1, GATE 2)
CV inputs	11
Gate inputs	9
Sequencer	16 songs × 4 parts × 32 steps, 1–128 beats per step
Presets	16 button presets
Display	128 × 64 OLED (emulated)

CREDITS

Design, engine and panel: betobullet. Behaviour modelled on the ACL Sinfonion user manual (firmware 1.0); Sinfonion is a product of Audiophile Circuits League and this is an independent recreation. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).